

High School Biology

Course Syllabus -- A Classical Liberal Arts Approach

36-Week Course | Florida NGSSS Aligned | Live and On-Demand

Course Philosophy

Biology is the study of life -- and you are alive. That is not a trivial observation. Every topic in this course connects directly to the experience of being a living human being: the food you eat, the air you breathe, the genes you inherited, the ecosystems you depend upon, and the evolutionary history you share with every living thing on Earth. This course is built on the conviction that biology is not a body of facts to be memorized for an exam but a way of seeing the world that, once learned, cannot be unlearned.

This course follows a classical liberal arts model. You will not only learn what biologists know -- you will learn how they came to know it, who first asked the questions, and why the answers matter for human life. The history of biology is a human story of careful observation, brilliant inference, stubborn error, and eventual revision. You will read and reason as scientists do, using the Claim-Evidence-Reasoning (CER) framework throughout the year to build arguments from evidence rather than accepting conclusions on authority.

The course moves from the microscopic to the macroscopic -- from atoms and molecules to cells, organisms, populations, ecosystems, and the full diversity of life on Earth. Each unit connects to the next. The connections are not incidental -- they are the architecture of biology itself.

This course accommodates both live (synchronous) instruction and on-demand (asynchronous) independent study. Each unit includes readings, laboratory activities or virtual lab alternatives, CER writing assignments, discussion prompts, and assessments clearly labeled for both formats.

Assessment Philosophy and Grading

Assessment in this course measures your ability to think biologically, not merely to recall terms. The following categories guide evaluation:

- CER Writing Assignments -- 30% -- Claim-Evidence-Reasoning responses to lab data, case studies, and biological questions
- Unit Assessments -- 25% -- Essays, short-answer tests, and concept checks, EOC-aligned
- Laboratory Work and Scientific Notebooks -- 20% -- Lab reports, observation journals, data analysis
- Discussion and Participation -- 15% -- Socratic seminars (live) or structured discussion board posts (on-demand)
- Midterm and Final Examinations -- 10% -- Cumulative, Florida Biology EOC-format

The CER Framework used in all written scientific reasoning:

- Claim -- A precise, testable answer to a scientific question
- Evidence -- Specific data, observations, or documented findings that support the claim
- Reasoning -- A logical explanation connecting the evidence to the claim using scientific principles

Vetted Online Resources for Students and Teachers

Open-Access Textbooks

- OpenStax Biology 2e (college-level, comprehensive, free):
<https://openstax.org/details/books/biology-2e>
- CK-12 Biology (high school level, free, adaptive): <https://www.ck12.org/biology/>
- Khan Academy AP Biology (free video lessons and practice sets):
<https://www.khanacademy.org/science/ap-biology>

Supplementary Science and Curriculum Resources

- HHMI BioInteractive (vetted videos, simulations, case studies, free):
<https://www.biointeractive.org/>
- Scitable by Nature Education (peer-reviewed science for students):
<https://www.nature.com/scitable/>
- NCBI PubMed and Genetics Resources: <https://www.ncbi.nlm.nih.gov/>
- PBS LearningMedia Florida (standards-aligned videos and activities):
<https://florida.pbslearningmedia.org/>

- National Geographic Education (ecology, biodiversity):
<https://education.nationalgeographic.org/>
- PhET Interactive Simulations, University of Colorado (biology and chemistry):
<https://phet.colorado.edu/>

Florida-Specific and EOC Preparation Resources

- CPALMS Florida Standards and Resources:
<https://www.cpalms.org/Public/search/Standard>
- Florida Museum of Natural History (biodiversity, ecology, evolution):
<https://www.floridamuseum.ufl.edu/>
- NSTA (National Science Teaching Association, teacher resources):
<https://www.nsta.org/>
- Visible Body Human Anatomy Atlas (human body systems, free browser version available): <https://www.visiblebody.com/>
- Genetics Science Learning Center, University of Utah:
<https://learn.genetics.utah.edu/>

UNIT 1 -- Nature of Science

Weeks 1-2 | Duration: 2 Weeks

Standards

SC.912.N.1.1, SC.912.N.1.3, SC.912.N.1.4, SC.912.N.1.6, SC.912.N.2.1, SC.912.N.2.2, SC.912.N.2.4, SC.912.N.3.1, SC.912.N.3.4

Unit Overview

This unit establishes the intellectual foundation for the entire course. Before students can study cells, genetics, or evolution, they must understand what science is, how it works, and what gives scientific claims their authority. Students who leave this unit knowing the difference between an observation and an inference, a theory and a guess, and scientific consensus and scientific certainty are equipped to reason honestly in every unit that follows. This is not housekeeping -- it is the grammar of biological thinking.

Historical Connections

Science as a method did not emerge fully formed. Students trace its development from Aristotle's natural philosophy, through the Islamic Golden Age (particularly Ibn

al-Haytham's formalization of experimental method around 1011 CE), to Francis Bacon's inductive reasoning in the 17th century, to Karl Popper's philosophy of falsifiability in the 20th century. The history of science is a history of revision -- a deeply human story of error corrected by evidence. Students examine how the practice of science has changed and why that change represents progress.

Human Connections

How do you evaluate a health claim on social media? How do you know whether a new medication works? How do you assess whether a news headline about a scientific study is accurate? These are not abstract questions -- they are decisions students will face throughout their lives as patients, citizens, and professionals. This unit teaches the habits of mind that make such decisions possible.

Connection to Unit 2

The tools built here -- controlled experiments, CER reasoning, peer review, the distinction between theory and law -- are used to evaluate every claim in every unit that follows. Unit 2 opens with the chemistry of life, and students will immediately apply their scientific reasoning skills to evaluate evidence about the properties of water and macromolecules.

Week	Content Focus and Standards	Weekly Assignments and Assessments
	Scientific method and experimental design. Observations vs. inferences. Variables: independent, dependent, controlled. Hypothesis formation. Qualitative vs. quantitative data. Introduction to CER writing structure. Safety and laboratory procedures. Standards: SC.912.N.1.1, SC.912.N.1.3, SC.912.N.1.4, SC.912.N.1.6	LIVE/ON-DEMAND: CER Practice 1 -- analyze a flawed scientific study; identify the error in reasoning and rewrite the conclusion. Observation Journal Entry: record 10 scientific observations from daily life; classify each as qualitative or quantitative. Lab: Controlled Experiment Design Activity -- design an experiment to test a biological question of student's choice; submit a written experimental plan. Discuss (live seminar or async post): Is science the only way to know something? Defend your position.

UNI T 2 - - Mac rom olec ules and Prop ertie s of Wat er Weeks 3-4 Duratio n: 2 Weeks Stand ards SC.912 .L.18.1, SC.912 .L.18.2, SC.912 .L.18.3, SC.912 .L.18.4, SC.912 .L.18.1 1,		
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SC.912
.P.8.7

Unit Overview

Life is chemistry. Before students can understand the cell, they must understand what cells are built from and what the medium they operate in -- water - - makes possible. This unit establishes the molecular

vocabulary of biology : the four macromolecules (carbohydrates, lipids, proteins, nucleic acids), the unique properties of water that make life possible, and the role of enzymes as biological catalysts. These are not isolated topics - they are the chemical foundation		
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tion of every biological process in the course. Historical Connections The story of biochemistry is one of science's great detective stories. Friedrich Miescher discovered nucleic acids in 1869 without knowing what they did -- a model of observ		
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ation precedi ng compr ehensi on by decade s. Linus Pauling 's work on protein structu re was so close to crackin g the structu re of DNA that it nearly redirec ted history. The slow 19th- century recogni tion that enzym es are protein s (not vital forces) represe nts one of the import		
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ant concep tual revisio ns in biology . The history of nutritio n scienc e -- includi ng early mistak es about dietary fat that shaped public health policy for 50 years -- illustrat es how scientif ic consen sus can be wrong and why revisio n is a strengt h, not a weakn ess.		
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Human Connections Every student in this course eats food. Food is macromolecules. Why does your body process a bowl of rice differently from a steak? Why is drinking only distilled water dangerous? Why do the enzymes in your stomach work		
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at body temperature but denature at high fever? What does the word 'organic' on a food label mean chemically? This unit turns the chemistry of life into a lens for understanding the body you live in.

Connection to Unit 3

The macromolecules

studied here directly become the components of the cell studied in Unit 3. Phospholipids form the cell membrane. Proteins serve as membrane channels, receptors, and enzymes. Nucleic acids carry genetic information. Without this unit, Unit 3 is a list of names. With it,		
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Unit 3 is an explan ation.		
Week UNIT 3 - - Cell s: Stru ctur e, Fun ctio n, Tran spor t, and Ener gy Weeks 5-8 Duratio n: 4 Weeks		

Standards SC.912 .L.14.1, SC.912 .L.14.2, SC.912 .L.14.3, SC.912 .L.14.4, SC.912 .L.14.5, SC.912 .L.14.6, SC.912 .L.18.7, SC.912 .L.18.8, SC.912 .L.18.9, SC.912 .L.18.1 0, SC.912 .P.10.1 Unit Overview The cell is the fundamental unit of life. In this unit students encounter		
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ter the cell not as a diagram to memorize but as an elegant, self-organizing system whose structure makes its function inevitable. Students study cell theory and its history, compare prokaryotic and eukaryotic cells, analyze organelle function, examine how		
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cells regulate what crosses the membrane, and investigate how cells capture and release energy through photosynthesis and cellular respiration. This unit builds entirely on the chemistry of Unit 2 and provides the biological context for the genetics that follows in Units		
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4 and 5.		
Historical Connections Cell theory accumulated over two centuries. Robert Hooke named cells in 1665 after observing cork through a primitive microscope. Antoni van Leeuwenhoek -- a Dutch draper with no formal scientific training		

who ground his own lenses -- becam e the first person to observ e living microo rganis ms in the 1670s. Schleid en and Schwa nn propos ed the cellular basis of all life in 1838- 1839. Rudolf Vircho w's 1855 maxim, Omnis cellula e cellula -- all cells from cells -- comple ted the		
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theory and ended the last serious defense of spontaneous generation. Lynn Margulis's 1967 endosymbiotic theory, initially dismissed, proposed that mitochondria and chloroplasts were once free-living prokaryotes -- now one of the best-supported ideas in evolutionary		
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biology . Each of these scienti sts represe nts a differe nt model of how scienc e advanc es. Human n Conn ectio ns Cancer is funda mentall y a cell cycle proble m. Unders tanding why antibio tics kill bacteri a but not human cells require s		
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unders tanding the structu ral differe nce betwee n prokary otic and eukary otic cells. Unders tanding why you feel tired, why food gives you energy, and why marath on runner s carboh ydrate- load the night before a race require s unders tanding cellular		
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respiration. This unit is not abstract -- every one of its topics has a direct application to your health and your life. Connection to Unit 4 The cell contains the DNA that Unit 4 explores. The cell cycle introduced here is revisited in Unit 4		
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	with the full molecular detail of mitosis and meiosis. The genome introduced here as the cell's blueprint becomes the central subject of genetics. Week		
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UNI T 4 - - Gen etics : Here dity, Cell Divis ion, and Patt erns of Inhe ritan ce			
Weeks 9-13 Duratio n: 5 Weeks			
Stand ards			
SC.912 .L.16.1, SC.912 .L.16.2,			

SC.912		
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SC.912		
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SC.912		
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7		
Unit		
Overview		
How is		
biological		
information		
copied,		
organized,		
and		
transmitted		
from		
cell to		
cell		
and		
from		
parent		

to offspring? This unit addresses one of the most profound questions in biology . Students study DNA structure and replication, the cell cycle and mitosis , meiosis and sexual reproduction, Mendelian inheritance, and the many ways inheritance patterns		
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s deviate from simple Mendelian predictions. This unit builds directly on the cell biology of Unit 3 and lays the molecular foundation for the protein synthesis and biotechnology content of Unit 5.		
Historical Connections		
Gregor Mendel, an Augusti		

nian friar in what is now the Czech Republ ic, perfor med meticul ous experi ments with pea plants in the 1860s and articul ated the laws of segreg ation and indepe ndent assort ment -- laws that were ignored for 35 years after his death and then redisco vered			
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simultaneously by three scientists in 1900. Walter Flemming first observed and described chromosomes during cell division in 1879. The discovery of DNA's double helix in 1953 by Watson and Crick – with essential, often uncredited contributions from Rosalind		
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Franklin and her X-ray crystallography data - is one of the most consequential and also most ethically complicated stories in modern science. Students should engage with both the science and the story.		
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	Human Connections You are the product of your parents' genetics -- but you are not determined by it. Understanding inheritance means understanding the difference between genotype and phenotype, between genetic risk and genetic			
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destiny . Understand ing mutations ons means understand ing why some cancer s run in familie s and others do not. Understand ing chrom osomal abnor malities s means engagi ng with conditi ons like Down syndro me using scientific ic accura cy rather than stigma. Geneti cs is the biology		
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Conn ectio n to Unit 5

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Unit 5.
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genes are expressed as proteins -- the bridge between genotype and phenotype that Unit 5 builds in full detail. The mutations studied at the end of this unit become the molecular inputs to protein synthesis and to evolution.

Week

	MID TER M REVI EW AND EXA MIN ATIO N Week 14 Duratio n: 1 Week This week is dedicat ed to cumula tive review of Units 1-4 and the midter m examin ation. The examin ation is formatt ed in	helicase, DNA polymerase, and ligase. How DNA structure provides a mechanism for inheritance. Standards: SC.912.L.16.3, SC.912.L.16.12		direction of synthesis. Historical Reading: guide annotation of Watson and Crick's 1953 Nature paper (one page, excerpted). Discussion Post/Seminar: What role did Rosalind Franklin play in the discovery of DNA structure, and what does her story tell us about how science credits its contributors?
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alignment with the Florida Biology End-of-Course (EOC) exam: multiple choice, data analysis items, and CER short-response questions.			
• Mid term Review Activity 1: Full - course concept map connection			

		ting at lea st one key ide a fro m eac h of Unit s 1- 4 wit h ann otat ed expl ana tion s of eac h con nec tion			
		• Mid ter m Rev iew Acti vity 2: EO C- styl e pra ctic			

	d topi c fro m Unit s 1- 4; con stru ct a full y cite d scie ntifi c arg um ent usi ng cou rse con tent				
	• Mid ter m Exa min atio n: Unit s 1- 4, EO C- alig ned for mat				

	, 50 multiple choice plus two CER R short-response items			
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UNIT 5 - Biot ech nolo gy: Tran scri ptio n, Tran slati on, and Gen e Expr essi on

Weeks 15-17 Duration n: 3 Weeks	
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Standards

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6

Unit Overview

If DNA is the blueprint, proteins are the building blocks. This unit covers the central dogma of molecular biology: how genetic information flows from DNA to mRNA to protein, and how

that flow is regulat ed and applied . Studen ts study transcri ption, translat ion, the univers al genetic code, and gene expres sion. The unit conclu des with biotech nology -- the deliber ate applica tion of molecu lar biology to human purpos es -- and the scientif ic,			
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medical,
environmental,
and ethical
questions
those
applications
raise.

Historical Connections

The
central
dogma
was
articulated by
Francis
Crick in
1958.

The
cracking of the
genetic
code in
the
early
1960s
by
Marshall
Nirenberg and
Har Gobind

	Khorana -- showing that the same three-base codons specify the same amino acids in virtually every organism on Earth -- was one of the decisive achievements of 20th-century science. The development of recombinant DNA technology in the 1970s by		
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Boyer and Cohen opened the biotechnology era. CRISPR-Cas9, developed into a precise gene-editing tool by Jennifer Doudna and Emmanuelle Charpentier (Nobel Prize 2020), represents the current frontier of biological technology and one of the most profound			
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Huma n Conn ectio ns

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gene --
a direct
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of recombinant DNA technology. Cancer immunotherapy, gene therapy for inherited diseases, forensic DNA analysis, agricultural GMOs, and personalized medicine all rest on the molecular mechanisms students study in this unit. After this unit,			
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biotechnology headlines are no longer opaque .

Connection to Unit 6

Mutations -- studied in Unit 4 -- are now understood at their full molecular depth: a base substitution change is a codon, which changes an amino acid, which changes a protein, which

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Evoluti
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Week

UNITS 6 - Evolution: Change Over Time				exception? (Retroviruses and reverse transcription) HIV as a case study with public health connections			
Weeks 18-21 Duration: 4 Weeks							
Standards							
SC.912.L.15.1, SC.912.L.15.2, SC.912.L.15.3, SC.912.L.15.8, SC.912.L.15.10, SC.912.L.15.13,							
SC.912.L.15.1, SC.912.L.15.13,							

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SC.912
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5

Unit Overview

Evolution is the unifying theory of biology.

Nothing in biology makes complete sense except in the light of evolution -- the formulation of geneticist

Theodosius Dobzhansky. This unit presents the

				and what the scientific evidence can and cannot address about the origin of life. The goal is rigorous, honest engagement with evidence -- the classical liberal arts aim of forming well-reasoned conclusions from careful examination of sources.				
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Historical Connections

Jean-Baptiste Lamarck proposed the first systematic theory of biological change in 1809 -- his mechanism was wrong, but his question was exactly right. Charles Darwin and Alfred Russel Wallace independently

		developed the theory of natural selection; their papers were read together at the Linnean Society in 1858 in one of science's most remarkable coincidences. Darwin's five-year Voyage of the Beagle (1831-1836) provided the systematic observation that grounded On			
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Human Connections

Antibio
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and it
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the
most
serious
public
health
threats
of the
21st
century

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Unders
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		natural selecti on explain s why you must finish the entire course of antibio tics your doctor prescri bes. It explain s why the flu vaccin e change s each year. It explain s why your immun e system and the immun e system of a chimpa nze are so similar. Evoluti on is			
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not an abstraction -- it is the explanation for how your body came to be the way it is and why disease organisms are always changing.

Connection to Unit 7

Classification is built on evolutionary relationships. The Linnaean system and modern

cladistics are both attempts to reflect the evolutionary history of organisms in the way we name and group them. You cannot understand why scientists classify organisms as they do without understanding evolution. Unit 7 is the organizational system that evolution

			makes necess ary.			
			Week			
			UNI T 7 - - Clas sific atio n, Taxo nom y, and the Dive rsity of Life			
			Weeks 22-23 Duratio n: 2 Weeks			
			Stand ards			
			SC.912 .L.7.1			

Unit Overview

Classification is a human act performed in response to the biological fact of diversity. The Linnaean system, the modern three-domain system, and cladistics are tools for organizing and understanding the relationships among living things.

This unit connects directly to evolution (all modern classification systems attempt to reflect evolutionary relationships) and sets the stage for the ecology unit by giving students the vocabulary to discuss the diversity of organisms in ecosystems.

Historical Connections

Carl Linnaeus developed hierarchical classification and binomial nomenclature in the 18th century based on observable physical characteristics -- a system that has proven remarkably durable. The discovery of

				evoluti onary relation ships repeat edly forced revisio ns. Carl Woese' s reclass ificatio n of life into three domain s in the 1970s, based on riboso mal RNA sequen ces rather than physic al charact eristics , was initially dismis sed and is now standa rd -- a model of				
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evidence overturning consensus. The ongoing debate about where to classify viruses illustrates that classification is an active, living scientific enterprise.

Human Connections

Why does precise naming matter?
Misidentifying a pathogen

				en can mean prescri bing the wrong antibio tic. Miside ntifying a specie s can mean applyin g the wrong pestici de. Legal protect ion under the Endang ered Specie s Act depend s on specie s-level identifi cation. The classifi cation skills of this unit are practic al scientif				
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Conn ectio n to Unit 8

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unit --

is a prerequisite for ecological study. When a food web names producers, consumers, and decomposers, those categories are built on the classification principles studied here.

Week

UNIT 8 - Ecology: Life in Context

Weeks
24-29 |
Duration:
6
Weeks

Standards

SC 912
.L 17.2,
SC 912
.L 17.4,
SC 912
.L 17.5,
SC 912
.L 17.8,
SC 912
.L 17.9,
SC 912
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Unit Overview

Ecology is biology at its largest scale -- the study of how organisms relate to each other and to their non-living environment. This unit integrates the entire course: organisms are made of cells (Unit 3), powered by

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Histor ical Conn ectio ns

Ernst
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ecolog y in 1866, the same decade Mendel was perfor ming his pea plant experi ments. Gilbert White's Natural History of Selbor ne (1789) is among the first works of ecolog y in the English langua ge -- natural history as simulta neoust y scientif ic and humani stic. Aldo	
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Leopold's *A Sand County Almanac* (1949) articulated the land ethic: the extension of moral consideration beyond the human community to the land itself. Rachel Carson's *Silent Spring* (1962) used evidence-based scientific argument to change public policy

on DDT -- one of the most consequential acts of scientific communication in the 20th century and a model of CER reasoning applied to public life.

Human Connections

Every meal you eat traces back to an ecosystem. Every breath involves the

				carbon cycle. Every specie s extincti on makes the ecologi cal comm unity less resilien t. Every acre of defores tation has conseq uences that circle back to human welfare . Ecolog y is a scientif ic descrip tion of the system s that suppor t human life. Unders tanding				
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it is not optional for an informed citizen of the 21st century. This unit is where biology becomes policy.

Connection to Unit 9

Plants are the primary producers that anchor nearly every terrestrial food web studied in this unit. Unit 9 examines the structure and

function of plants in detail -
- explaining how the photosynthesis that drives every ecosystem actually occurs in a living organism. Ecology gives students the why; plant biology gives them the how.

Week

UNIT 9 - Plant Structure and Function Weeks 30-31 Duration: 2 Weeks Standards SC.912.L.14.7, SC.912.L.14.2 6, SC.912.L.14.2 7 Unit Overview			and Classification Activity. Species Interaction Study: identify interaction types in a coral community.
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Histor ical Conn ectio ns

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Father
of
Botany,
wrote

systematic works on plants in the 4th century BCE. Nehemiah Grew and Marcello Malpighi produced the first systematic plant anatomy in the 17th century. Stephen Hales in the 18th century first quantified water movement through plants.

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Gregor
Mendel
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genetic
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t who
spent
years
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g plant
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ization.
Agricul-
ture --
the
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onal
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ion of
plants -
- is

arguably the most consequential human application of biological knowledge in history.

Human Connections

Every grain, vegetable, and fruit you eat is a plant structure performing a biological function. Seeds are embryos with stored endosp

				erm; fruits are ripened ovary walls evolve d to dispers e seeds; leaves are solar panels perfor ming photos ynthesi s; roots are both anchor s and absorp tion organs. Approxi mately 80% of the human food supply depend s on floweri ng plants. Crop scienc e, food securit				
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y, deforestation policy, and medicinal plant development all require understanding plant biology.

Connection to Unit 10

The transition from plants to the human body reinforces one of the deepest insights in biology: the unity of

[illegible]

machinery,
ultimately
shaped
by the
same
evolutionary
history.

Week

UNIT 10

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Human Body Systems

Weeks
32-35 |
Duration:
4

Weeks

Standards

SC.912.
L.14.2
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Unit Overview

The human body is the most directly relevant biological system students will ever study -- because it is theirs. This final content unit brings together the entire course: the chemistry of macro molecu

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the evolution of Unit 6 explains why human anatomy looks the way it does. Homeostasis - the maintenance of stable internal conditions -- serves as the integrating theme of the unit.

Historical Connections

The history of human anat

my is a story of courage overcoming institutional tradition. Andreas Vesalius (1543) revolutionized anatomy by conducting his own dissections at a time when Galenic medicine, 1,400 years old, was still considered authoritative. William Harvey (1628) describ

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Human Connections

This is the most personal unit in the course. Cardiovascular disease is the leading cause of death in the United States - understanding how the heart and blood vessels work is literally life-saving knowledge. Understanding

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Unit 5.
The
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Connection Forward

This
unit is
the
culmination
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course.
Students
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followed
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progression
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atoms
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molecules to cells to genetics to evolution to ecosystems to plants now arrive at the organism level of their own biology. The year ends not with the cosmos but with the body -- the most intimate expression of the biological principles studied all year.

FINAL REVIEW AND EXAMINATION

Week 36 | Duration: 1 Week
The final week is dedicated to cumulative review of all 10 units and the final examination, formatted in alignment with the Florida

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	when the evidence warrants it. Biology is the study of life. You are alive. The two facts are not unrelated. Week		
Week 32		Organization of the human body: cells, tissues (epithelial, connective, muscle, nervous -- structure and function of each), organs, organ systems, organism. Homeostasis: concept, importance, negative feedback loops (body temperature, blood glucose), positive feedback loops (childbirth, blood clotting). Integumentary system: skin layers, functions (protection, thermoregulation, sensation, synthesis of vitamin D). Skeletal and muscular systems: bone composition and functions, types of joints, skeletal vs. smooth vs. cardiac muscle, mechanism of muscle contraction overview. Standards: SC.912.L.14.26	LIVE/ON-DEMAND: Body Organization Lab -- construct a hierarchy from cell to organ system; provide annotated examples from multiple systems; identify tissue types at each level. Homeostasis Conceptual Change Experiment (CER): analyze a clinical scenario (fever, heat stroke) and explain the feedback mechanism that restore homeostasis, identifying what is disrupted and how the body responds. Tissue Type Lab (microscopy slides or virtual images of human Body) -- identify, sketch, and explain the four tissue types. Homeostasis Conceptual Change Experiment: all major homeostatic variables (temperature, sugar, blood pressure, pH) with their sensor, control center, and effector.
Week 33		Digestive system: mouth to anus, enzymes and digestion (amylase, pepsin, lipase), small intestine absorption, role of liver and pancreas, connection to macromolecules (Unit 2). Circulatory system: heart anatomy and cardiac cycle, pulmonary vs. systemic circulation, blood vessels (arteries, veins, capillaries), blood components (red and white blood cells, platelets, plasma), blood pressure, cardiovascular disease as a biological and public health problem. Respiratory system: gas exchange at alveoli, breathing mechanics, connection to cellular respiration (Unit 3). Standards: SC.912.L.14.26, SC.912.L.14.36	LIVE/ON-DEMAND: Heart Rate Lab -- measure pre and post-exercise heart rate and blood pressure; or analyze provided data set (async); write conclusions of cardiovascular fitness. Enzyme Activity Lab: test the effect of pH and temperature on amylase or pepsin activity; graph results; write conclusions connecting results to optimal digestive enzyme activity. Cardiovascular Disease Case Study: read a clinical scenario; identify risk factors; interpret lab values; recommend and justify lifestyle interventions using evidence. Annotate a diagram to trace the path of one glucose molecule from ingestion/inhalation to ATP production in a muscle cell -- a synthesis of Units 2 and 3.

Week 34	Excretory system: kidney anatomy, nephron structure and filtration/reabsorption/secretion steps, urine formation, osmotic regulation of blood composition. Nervous system: neuron structure (dendrites, cell body, axon, myelin sheath), action potential mechanism, synapse and neurotransmitter function, central nervous system (brain, spinal cord) vs. peripheral nervous system, reflex arcs. Endocrine system: major glands and hormones (insulin, glucagon, adrenaline, thyroid hormone, sex hormones), target organ specificity, comparison of hormonal vs. nervous signaling in speed and duration. Standards: SC.912.L.14.26	LIVE/ON-DEMAND: Neuron Structure and Action Potential Diagram -- annotate neuron anatomy and the electrical and chemical events of an action potential from resting potential through depolarization, repolarization, and synaptic transmission. Nervous System vs. Endocrine System Comparison Chart -- compare the two systems in terms of mechanism, speed, duration, examples, and disorders for each. Hormonal Disorder Lab -- choose one disorder (Type 1 diabetes, hypothyroidism, Cushing syndrome); explain the biological basis and the disrupted feedback loop; discuss treatment options using evidence. CER Short Essay: How does the excretory system maintain blood homeostasis? Trace a urea molecule from formation to excretion.
Week 35	Immune system: first line of defense (skin and mucous membranes), second line of defense (non-specific: phagocytes, inflammatory response, fever, complement proteins), third line of defense (specific adaptive immunity: B lymphocytes and antibody production, T lymphocytes and cell-mediated immunity, antigen presentation, clonal selection, immunological memory). Vaccines: mechanism of action, herd immunity, history of vaccination from Jenner to mRNA vaccines. Antibiotics vs. antiviral drugs: why the molecular distinction between prokaryotic and eukaryotic cells (Unit 3) determines treatment. Autoimmune diseases and allergies as immune system dysregulation. Standards: SC.912.L.14.52	LIVE/ON-DEMAND: Immune Response Diagram -- trace the full immune response from infection from first contact through B cell activation, antibody production, and memory cell formation. Vaccine Mechanism Diagram -- annotate each step. Vaccine Safety Debate -- annotate how a vaccine stimulates immunity without causing disease; include mRNA vaccine as a modern example (connecting to transcription/translation). CER Summative Essay: Evaluate the scientific evidence for vaccine safety and efficacy -- construct an argument using epidemiological data, molecular action, and historical evidence. Unit 10 Connection Reflection: Select one concept from the immune system and trace its application through at least two subsequent units -- demonstrate the coherence of the entire course.
Week 30		
Week 31	Plant vascular system in detail: xylem structure (tracheids, vessel elements) and the cohesion-tension mechanism of water transport. Phloem structure (sieve tubes, companion cells) and the pressure-flow hypothesis of sugar transport. Transpiration: process, measurement, and role in driving xylem transport. Plant reproduction: pollination mechanisms, fertilization (double fertilization in angiosperms), seed structure and germination, fruit development, asexual reproduction. Plant hormones: auxin (phototropism, gravitropism), gibberellins (stem elongation), ethylene (fruit ripening). Tropisms. Standards: SC.912.L.14.27	LIVE/ON-DEMAND: Transpiration Lab (live or virtual) -- plastic bag method, or virtual simulation to measure water loss under varying conditions (light intensity, humidity, wind); graph and analyze data. Vascular Transport Diagram -- trace the complete pathway of water from roots to leaves through stomata and sugar from mesophyll cells to roots, annotating the mechanism driving each movement. Unit 9 Assessment: short-answer and diagram labeling. Connection Reflection: How does the plant vascular system compare structurally and functionally to the human circulatory system? What study in Unit 10? What does this comparison tell us about convergent solutions to the same problem?
Week 24		

Week 25	Biomes: terrestrial biomes (tropical rainforest, temperate deciduous forest, grassland, chaparral, desert, taiga, tundra) and aquatic biomes (freshwater: lakes, rivers, wetlands; marine: open ocean, intertidal, coral reef; estuarine). Climate as the primary determinant of biome distribution. Comparison of biodiversity across biomes. Standards: SC.912.L.17.2	LIVE/ON-DEMAND: Biome Climate Graph (climatograph) Analysis -- interpret precipitation and temperature graphs for 6 biomes; identify climate zones using CER reasoning; explain why organisms in each biome have particular adaptations. Biome Table: abiotic conditions, dominant plant and animal, representative animals, current threats, and human connection for each. Reading and discussion excerpt from Aldo Leopold's A Sand County. Discussion Post/Seminar: Does a biome benefit from organisms that live in it, or to the humans? What framework should govern human development in wild land?
Week 26	Energy flow in ecosystems: producers, primary consumers, secondary consumers, tertiary consumers, decomposers and detritivores. Food chains and food webs. Trophic levels. Energy pyramids: the 10% rule and energy loss between trophic levels. Biomagnification: how toxins concentrate at higher trophic levels. Keystone species and their disproportionate ecological roles. Standards: SC.912.L.17.9	LIVE/ON-DEMAND: Food Web Construction -- build a complex food web for a Florida ecosystem; analyze consequences of removing a key species using CER reasoning. Energy Pyramid Question -- calculate energy available at each trophic level from producer energy value; graph results; explain using CER. Biomagnification Case Study: trace a persistent toxin (DDT or mercury) through trophic levels; explain why top predators accumulate the highest concentrations. Human Diet and Trophic Level Discussion: What are the ecological implications of a meat-heavy diet vs. a plant-based diet? Final Ecosystem Project first draft due.
Week 27	Biogeochemical cycles: carbon cycle (including photosynthesis, cellular respiration, decomposition, combustion, and ocean absorption), nitrogen cycle (fixation, nitrification, assimilation, denitrification), water (hydrologic) cycle, phosphorus cycle. Human disruption of biogeochemical cycles: fossil fuel combustion and carbon, synthetic nitrogen fertilizers and eutrophication. Standards: SC.912.L.17.5	LIVE/ON-DEMAND: Carbon and Nitrogen Cycle Diagrams -- Annotated Diagrams -- construct complete diagrams with human impact annotations at each step. CER Short Essay: Trace a carbon atom from atmospheric CO₂ through photosynthesis, a food chain, decomposition, and return to the atmosphere -- identify at which point humans have altered each step. Eutrophication Case Study -- analyze water quality data from a Florida water body affected by agricultural runoff; write a CER recommendation for remediation. Florida Research Project presentations due. Quiz on biogeochemical cycles and energy flow.
Week 28	Population ecology: exponential vs. logistic growth models, J-curve and S-curve, carrying capacity, limiting factors (biotic: predation, competition, disease; abiotic: temperature, water, nutrients). Demographic analysis: births, deaths, immigration, emigration. Predator-prey population oscillations. Ecological succession: primary succession (bare rock to climax community) and secondary succession (disturbed ecosystem recovery). Climate change and ecosystem dynamics. Standards: SC.912.L.17.5, SC.912.L.17.4	LIVE/ON-DEMAND: Population Growth Graph -- plot and compare exponential and logistic growth curves; calculate carrying capacity; identify limiting factors using CER analysis. Predator-Prey Activity (physical or digital) -- model population oscillations in a lynx-hare system; explain population ecology principles. Succession -- compare primary succession on a volcanic island (Surtsey) with secondary succession following fire; annotate diagrams with time scale and environmental changes. Quiz: population ecology and succession.

Week 29	Biodiversity: genetic, species, and ecosystem diversity and why each level matters for ecosystem stability. Causes of biodiversity loss: habitat destruction, climate change, overexploitation, pollution, invasive species. Invasive species in Florida: Burmese pythons, lionfish, Brazilian pepper, air potato -- mechanisms of impact. Human impact: deforestation, urban sprawl, fossil fuel combustion. Renewable vs. nonrenewable resources: costs, benefits, trade-offs. Conservation biology and environmental policy. Standards: SC.912.L.17.4, SC.912.L.17.8, SC.912.L.17.11, SC.912.L.17.13, SC.912.L.17.16, SC.912.L.17.20	LIVE/ON-DEMAND: Florida Invasive Species Research and Presentation -- analyze the evidence for the impact of one Florida invasive species; evaluate proposed management strategies; structured argument for the best policy recommendation. Silent Spring CER Analysis: evaluate Rachel Carson's scientific argument against DDT using the CER framework -- identify her claim, the evidence, and the quality of her reasoning. Unit 8 Test: choice, food web and data analysis, and connection on human-ecosystem conflict. Connection Reflection: How does understanding natural selection help explain why invasive species are often so successful in their new environments?
Week 22		
Week 23	Cladistics and phylogenetic trees: shared derived characteristics, outgroups, interpreting branching diagrams, monophyletic groups. Molecular phylogenetics: using DNA and protein sequence data to establish evolutionary relationships (connection to Unit 6 molecular clocks). Comparing Linnaean and cladistic classification approaches. Characteristics and ecological roles of major kingdoms. Viruses: structure, diversity, and the ongoing classification debate. Standards: SC.912.L.7.1	LIVE/ON-DEMAND: Cladogram Construction -- build a cladogram for a set of organisms (invertebrates, bacteria, fungi, plants, animals, and a protist) using a matrix of shared derived characteristics; analyze and interpret evolutionary relationships. Kingdom Comparison Chart: cell type, reproduction, metabolism, ecological role, and one human-relevant application for each kingdom. Short Essay: Are viruses alive? Use the characteristics of life and the principles of classification to support a reasoned, evidence-based argument. Unit 7 Assessment: taxonomy and classification of life. Connection Reflection: How does the molecular evidence for classification (ribosomal RNA) support the unity of the content across Units 5, 6, and 7?
Week 18		
Week 19	Natural selection: required conditions (heritable variation, differential reproductive success, selection pressure). Darwin and Wallace's contributions. Types of natural selection: directional, stabilizing, disruptive. Adaptation. Artificial selection and its relationship to natural selection. Real-time example: antibiotic resistance as natural selection in bacterial populations. Standards: SC.912.L.15.13, SC.912.L.15.3	LIVE/ON-DEMAND: Antibiotic Resistance Simulation (HHMI BioInteractive -- Evolution of Drug Resistance) -- run the simulation; write a full CER analysis of the results using natural selection principles. Moth Case Study: analyze population data before and after industrial melanism; graph allele frequency changes; write CER conclusion. Reading: excerpt from Darwin's On the Origin of Species, Chapter 4 (Natural Selection) with guided annotation questions. Discussion/Seminar: What is the difference between an individual adapting and a population evolving? How does this distinction matter?
Week 20	Mechanisms of evolutionary change beyond natural selection: genetic drift (founder effect, population bottleneck), gene flow, sexual selection. How mutation and genetic recombination generate heritable variation on which natural selection acts. Molecular clocks: using rates of DNA sequence change to estimate	LIVE/ON-DEMAND: Genetic Drift Simulation (beads or digital) -- model founder effect and bottleneck effect; analyze how small population size accelerates allele frequency change independent of natural selection. Molecular Clock Activity: use DNA sequence data from multiple species to estimate divergence times.

				divergence time. Phylogenetic trees and cladograms: how evolutionary relationships are represented. Speciation: reproductive isolation mechanisms. Standards: SC.912.L.15.14, SC.912.L.15.15, SC.912.L.15.2, SC.912.L.15.3				and construct a simple phylogenetic tree; interpret results using CER. Mechanisms of Evolution: Comparison Chart: mechanism, definition, effect on allele frequency, real-world example of natural selection and other evolutionary mechanisms.
	Week 21			Scientific explanations for the origin of life: geological timeline of early Earth, chemical evolution hypothesis, Miller-Urey experiment (1953) and its results and limitations, RNA World hypothesis, formation of protocells. What these hypotheses explain and what remains genuinely open. Basic trends in hominid evolution: fossil record from early Australopithecus to Homo sapiens, key anatomical changes, timeline. Standards: SC.912.L.15.8, SC.912.L.15.10				LIVE/ON-DEMAND: Origins of Life Scientific Evaluation -- analyze the Miller-Urey experiment, the RNA World hypothesis as scientific models. CER criteria: What do they explain? What evidence supports them? What do they not explain? What questions remain open? Human Evolution Timeline Project: construct an annotated timeline of human evolution using fossil evidence from at least 5 species; include key anatomical and behavioral changes. Unit 6 Test: multiple choice, data analysis, and CER essay. Connection Reflection: How does molecular evidence for evolution (DNA sequence comparisons) connect to what you learned about replication in Unit 4 and protein synthesis in Unit 5?
	Week 15							
	Week 16			Translation: ribosome structure (large and small subunits), mRNA-ribosome interaction, codons and anticodons, aminoacyl-tRNA, peptide bond formation, polypeptide elongation, start codon (AUG), stop codons (UAA, UAG, UGA). The universal genetic code: codon chart, degeneracy, universality. Protein folding: primary, secondary, tertiary, quaternary structure. How point mutations alter amino acid sequences and protein function. Standards: SC.912.L.16.13, SC.912.L.16.9				LIVE/ON-DEMAND: Translation Simulation Activity: use the codon chart to translate an mRNA sequence into an amino acid chain; then introduce a point mutation and show how the amino acid sequence changes. Sickle Cell Disease CER Case Study: use genetic code to trace the molecular path from a base substitution in DNA to the abnormal hemoglobin protein to the disease phenotype. Protein Synthesis Complete Flow Chart: DNA to mRNA to protein; annotations at every step. Genetics Learning Center (University of Utah) Translation Module (on-demand) Protein Synthesis Quiz.
	Week 17			Biotechnology: restriction enzymes, recombinant DNA, plasmids and vectors, bacterial transformation. PCR (polymerase chain reaction): steps and applications. Gel electrophoresis: how it works, how to read results, forensic and medical uses. DNA sequencing overview. CRISPR-Cas9: mechanism and current applications. GMOs in agriculture and medicine. Forensic DNA analysis and paternity testing. Ethical, legal, and social implications. Standards: SC.912.L.16.10				LIVE/ON-DEMAND: Gel Electrophoresis Simulation (physical or virtual via Learn.Genetics or HHMI) separate and compare DNA fragment patterns. CER analysis identifying a suspect or diagnosing a patient. Biotechnology Ethics Research Project: choose one application (CRISPR gene therapy, agricultural GMO, forensic DNA database, direct-to-consumer genetics); research the science, benefits, risks, and ethical dimensions; present as a structured CER argument. Socratic Seminar (live) or structured Discussion Post (async): Should humans edit the human germline? What limits, if any, should exist? 5 Assessment: short-answer and CER essay.
	Week 9							
	Week 10			Cell cycle in detail: G1 (growth, protein synthesis), S phase (DNA replication), G2 (preparation for division), M phase (mitosis and cytokinesis). Mitosis stages: prophase, metaphase, anaphase, telophase.				LIVE/ON-DEMAND: Onion Root Tip Mitosis Lab (microscope or virtual microscopy via HHMI) -- observe and count cells in each mitotic stage; calculate relative time spent in each phase; full CER analysis.

		Chromosome behavior during mitosis. Role of cell division in growth, repair, and reproduction. Cell cycle checkpoints and cancer as a cell cycle regulatory failure. Standards: SC.912.L.16.14, SC.912.L.14.5		Cancer Case Study CER: use the cell cycle to explain why cancer cells divide uncontrollably -- identify a checkpoint that fails in a given cancer type. Cell Cycle Annotation Diagram: label each phase with key molecular events. Quiz: DNA structure, replication, cell cycle.
Week 11		Meiosis: overview and purpose, meiosis I (homologous chromosome separation) and meiosis II (sister chromatid separation), crossing over and genetic recombination during prophase I, independent assortment of chromosomes. Fertilization: gamete formation, haploid and diploid chromosome numbers. Comparison of mitosis and meiosis: purpose, number of divisions, daughter cells, genetic outcome. Standards: SC.912.L.16.15, SC.912.L.16.16, SC.912.L.16.17		LIVE/ON-DEMAND: Mitosis vs. Meiosis Comparison Chart -- side-by-side analysis of purpose, number of divisions, chromosome number of daughter cells, genetic variation produced. Meiosis Simulation (pipe cleaner chromosomes or virtual simulation) model independent assortment and crossing over, record outcomes. CER Short Essay: Why does sexual reproduction produce more genetic variation than asexual reproduction, and what is the evolutionary significance of that variation (preview of Unit 6)? Fertilization Diagram: annotate the pathway from meiosis through gamete formation to fertilization.
Week 12		Mendelian genetics: law of segregation and law of independent assortment. Monohybrid and dihybrid crosses. Punnett squares. Genotype, phenotype, homozygous, heterozygous, dominant, recessive. Test crosses. Probability in genetics. Sex determination in humans: X and Y chromosomes. Sex-linked traits and their inheritance patterns. Non-Mendelian inheritance: incomplete dominance, codominance, polygenic inheritance, multiple alleles (ABO blood type). Standards: SC.912.L.16.1, SC.912.L.16.2, SC.912.L.16.5, SC.912.L.16.9		LIVE/ON-DEMAND: Punnett Square Problem Set: problems including monohybrid, dihybrid, sex-linked, and codominance crosses; each answer accompanied by a CER written explanation of the reasoning, not just the ratio. Genetics Word Problems with real-world medical contexts (blood type compatibility, hemophilia, colorblindness, sickle cell trait). Family Pedigree Analysis: trace a genetic trait through three generations and determine the inheritance pattern. Mendel Historical Reflection: one-page essay -- why was Mendel's work ignored for 35 years, and what factors that tell us about how scientific communities evaluate new ideas?
Week 13		Mutations: types (point mutations -- substitution, insertion, deletion; chromosomal mutations -- deletion, duplication, inversion, translocation). Effects of mutations: silent, missense, nonsense, frameshift. Causes of mutation: replication errors, mutagens (radiation, chemicals), viruses. Chromosomal abnormalities: trisomy 21 (Down syndrome), Turner syndrome (45,X), Klinefelter syndrome (47,XXY). Genetic disorders caused by mutations. Standards: SC.912.L.16.4, SC.912.L.16.2		LIVE/ON-DEMAND: Karyotype Analysis Activity: identify chromosomal abnormalities from karyotype images; write a CER diagnostic explanation for a case. Mutation Types Illustrated Chart: classify, and give a real biological example of each mutation type. Genetic Disorder Research Brief: choose a genetic disorder, research its molecular basis (what gene, what mutation, what protein is affected), explain the connection from mutation to phenotype. Unit 4 Test: multiple choice, Punnett square problem, data analysis, and one CER short essay. Connection Reflection: How will a mutation in a protein-coding gene affect the protein produced in Unit 5?
Week 5				
Week 6		Plant and animal cell organelles: nucleus, mitochondria, chloroplast, rough and smooth ER, Golgi apparatus, ribosomes, lysosomes, vacuoles, cell wall, centrioles. Structure-to-function analysis for each organelle. Cell specialization: how cells differentiate in the human body (muscle cells, nerve cells, red blood cells). Comparison of plant and animal cell structures.		LIVE/ON-DEMAND: Cell Organelle Model Project (physical 3D model or digital illustration) -- build and label a cell with function annotations for each organelle; include a comparison of what is present in plant cells vs. animal cells only. Organelle Analogy Essay: compare the cell to a human institution (company, factory, school); assign each organelle to a

	Standards: SC.912.L.14.2, SC.912.L.14.3	corresponding role and explain the structural part. Short Quiz: organelle identification and function. Cell Structure Interactive Activity (on-demand).
Week 7	Cell membrane: fluid mosaic model, phospholipid bilayer, membrane proteins (channel proteins, carrier proteins, receptor proteins). Passive transport: simple diffusion and osmosis (including hypotonic, hypertonic, isotonic solutions). Facilitated diffusion. Active transport and energy requirements. Cell cycle overview: interphase (G1, S, G2) and role of cell division in growth and tissue repair. Standards: SC.912.L.14.6, SC.912.L.14.5	LIVE/ON-DEMAND: Osmosis Lab (potato cube or osmosis in salt solutions of varying concentrations) measure mass change; create a data table and graph; write full CER lab report. Virtual alternative: Osmosis and Diffusion simulation via PhET or CK-12. Cell Transport Diagram: illustrate all transport types with directional arrows, energy requirements, and real biological examples. Medical Application CER: Why must IV saline solution be isotonic? What would happen to red blood cells in a hypotonic or hypertonic IV solution? Quiz: cell membrane structure and transport.
Week 8	Photosynthesis: reactants and products, light-dependent reactions (chlorophyll, thylakoids, ATP, NADPH), light-independent reactions (Calvin cycle, glucose), overall equation. Cellular respiration: glycolysis, Krebs cycle, electron transport chain, ATP yield, aerobic vs. anaerobic pathways, fermentation. ATP as the cell's energy currency. Interrelationship of photosynthesis and cellular respiration. Energy transformations between forms. Standards: SC.912.L.18.7, SC.912.L.18.8, SC.912.L.18.9, SC.912.L.18.10, SC.912.P.10.1	LIVE/ON-DEMAND: Leaf Disk Photosynthesis Lab (spinach leaf disk floating assay under varying light intensities) or virtual equivalent (HHMI BioInteractive photosynthesis simulation) -- full CER lab report. Fermentation Lab: measure CO ₂ production under aerobic vs. anaerobic conditions with different sugar sources; CER analysis. Photosynthesis and Respiration Comparison Chart: inputs, outputs, location in cell, energy role, net ATP. Unit 3 Test: multiple choice, diagram labeling, and one CER short essay. Connection Reflection: Trace a carbon atom from atmosphere through photosynthesis, into a glucose molecule, then through cellular respiration to an ATP molecule. Identify every transformation.
Week 3		
Week 4	The four macromolecules -- carbohydrates: monosaccharides and polysaccharides, energy storage and structural roles. Lipids: fats, phospholipids, steroids, roles in membranes and energy. Proteins: amino acids, peptide bonds, levels of protein structure, functional diversity. Nucleic acids: DNA vs. RNA, nucleotide structure. Enzymes: structure, active site, substrate specificity, activation energy, denaturation. Standards: SC.912.L.18.1, SC.912.L.18.3, SC.912.L.18.4, SC.912.L.18.11	LIVE/ON-DEMAND: Macromolecule Testing Lab (physical or virtual) -- Benedict test, Lugol iodine test, Biuret test, Sudan IV test -- test unknown food samples; full CER lab report with data table, graph, conclusion. Food Label Analysis: identify and classify macromolecules in a real food product; explain each macromolecule's biological role. Unit 2 Assessment: short-answer test with diagram labeling. Connection Reflection: How does the phospholipid structure studied this week make the cell membrane of Unit 1 possible?
Week 1		
Week 2	Scientific theories vs. scientific laws -- how they differ and why a theory is not a guess. Peer review and the reproducibility of scientific knowledge. Scientific models and their limitations. History of the scientific method: Aristotle, Ibn al-Haytham, Bacon, Popper. Science and pseudoscience: how to tell the difference. Standards: SC.912.N.2.1, SC.912.N.2.2, SC.912.N.2.4, SC.912.N.3.1, SC.912.N.3.4	LIVE/ON-DEMAND: CER Essay -- evaluate a historical scientific claim (spontaneous generation) using the CER framework: was it good science for its time? What evidence overturned it? Historical Reading: guided annotation of an excerpt from Francis Bacon's <i>Novum Organum</i> (1620). Unit 1 Assessment: scientific reasoning quiz (multiple choice and short answer, EOC-format). Socratic Seminar (live) or Discussion Post (async): What makes a scientific claim trustworthy? Bring one example from current news.

